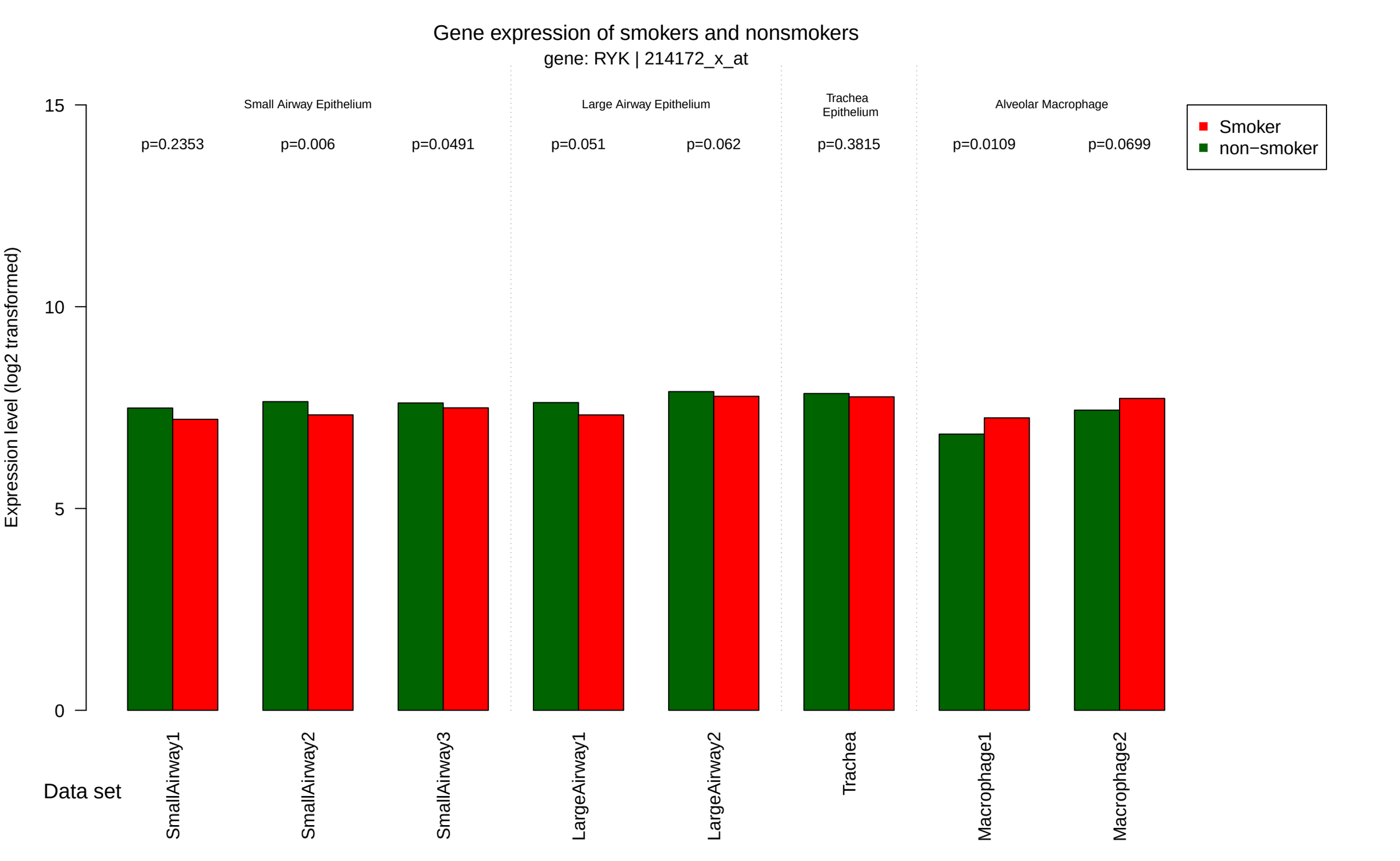
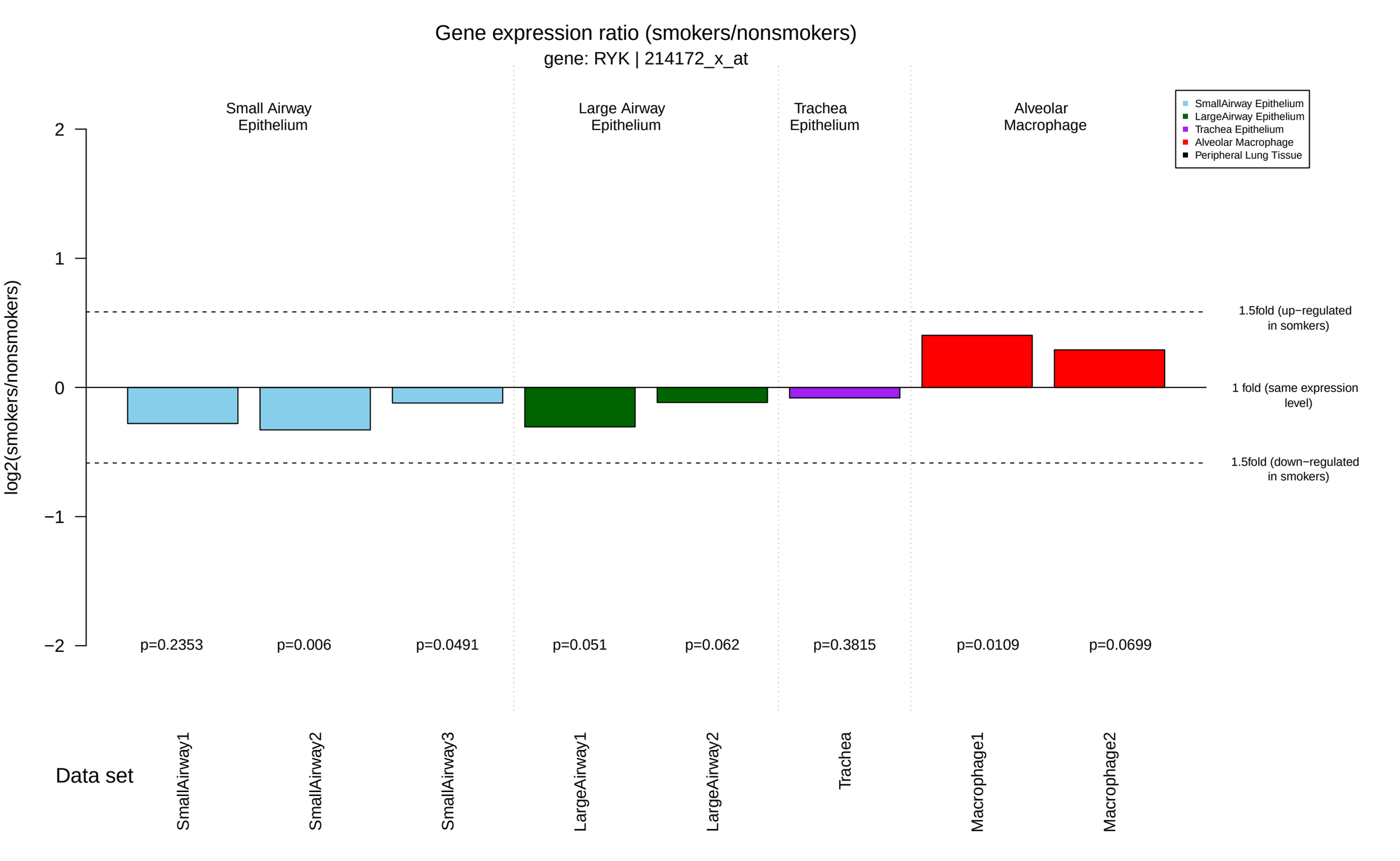
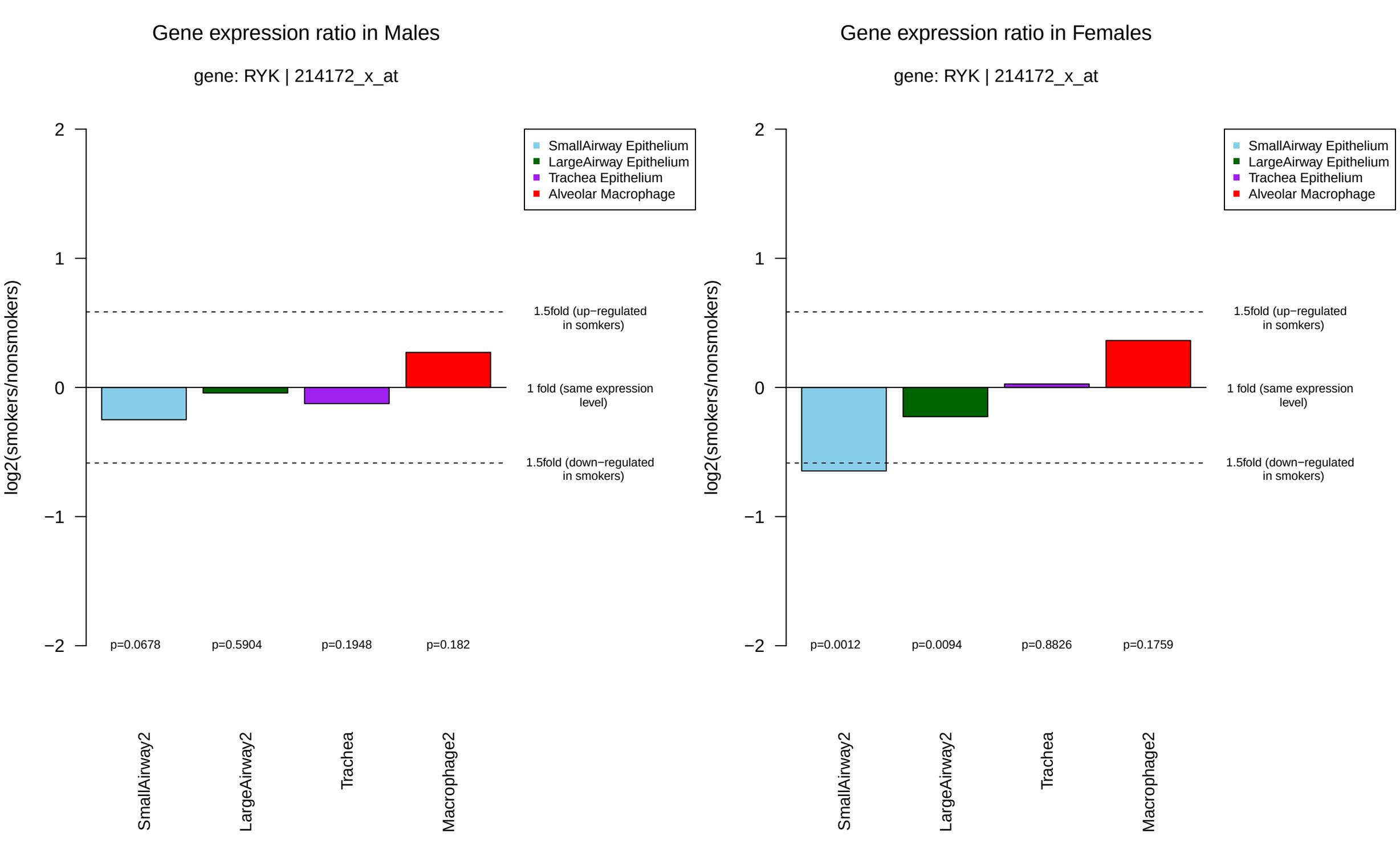
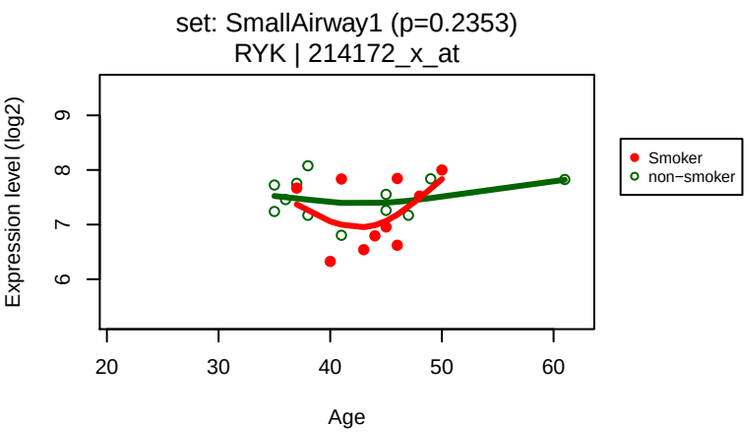
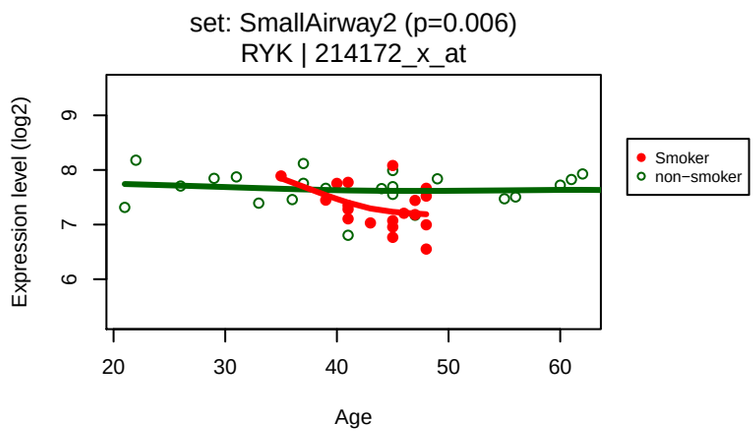
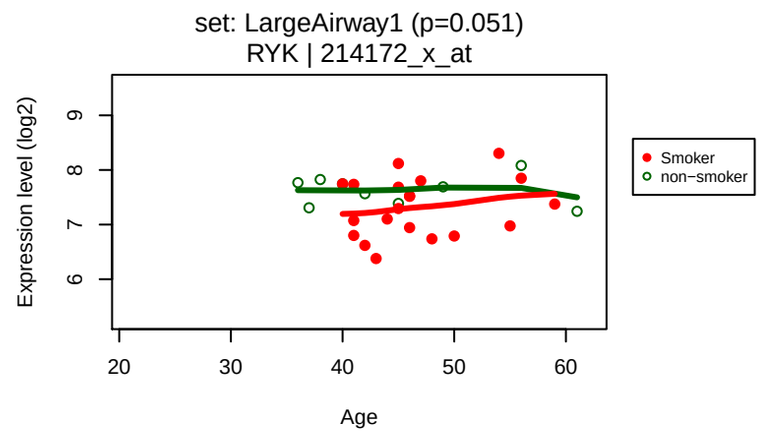

Gene expression of smokers and nonsmokers
gene: RYK | 214172_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2353
p=0.006
p=0.0491
p=0.051
p=0.062
p=0.3815
p=0.0109
p=0.0699
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

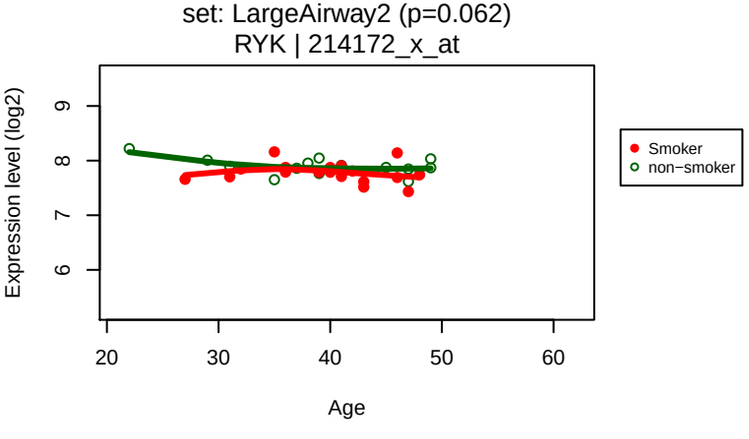

Gene expression ratio (smokers/nonsmokers)
gene: RYK | 214172_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2353
p=0.006
p=0.0491
p=0.051
p=0.062
p=0.3815
p=0.0109
p=0.0699
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

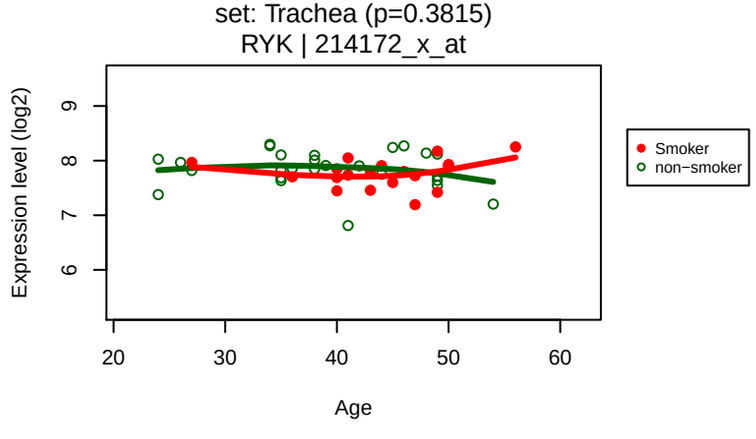

Gene expression ratio in Males
gene: RYK | 214172_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0678
p=0.5904
p=0.1948
p=0.182
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RYK | 214172_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0012
p=0.0094
p=0.8826
p=0.1759
SmallAirway2
LargeAirway2
Trachea
Macrophage2

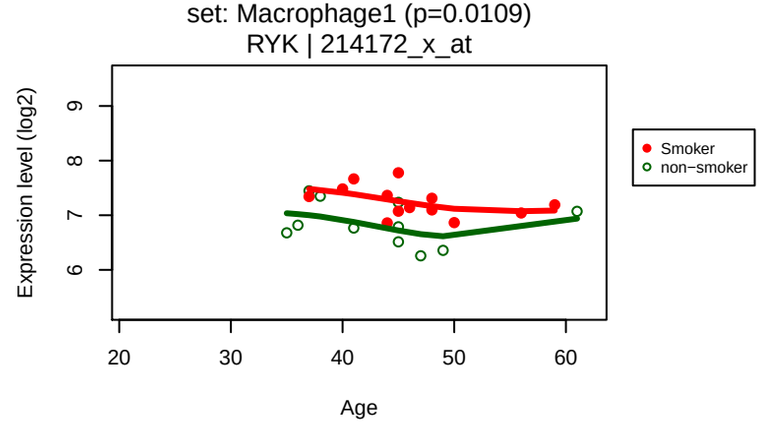

set: SmallAirway1 (p=0.2353)
RYK | 214172_x_at
Expression level (log2)
Age
Smoker
non−smoker

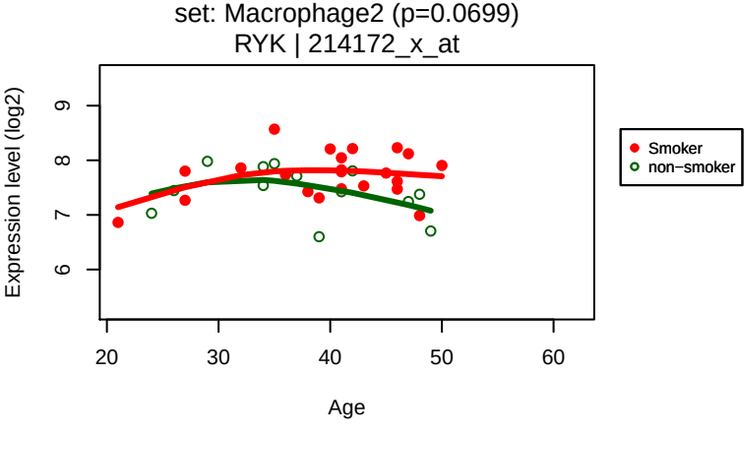

set: SmallAirway2 (p=0.006)
RYK | 214172_x_at
Expression level (log2)
Age
Smoker
non−smoker

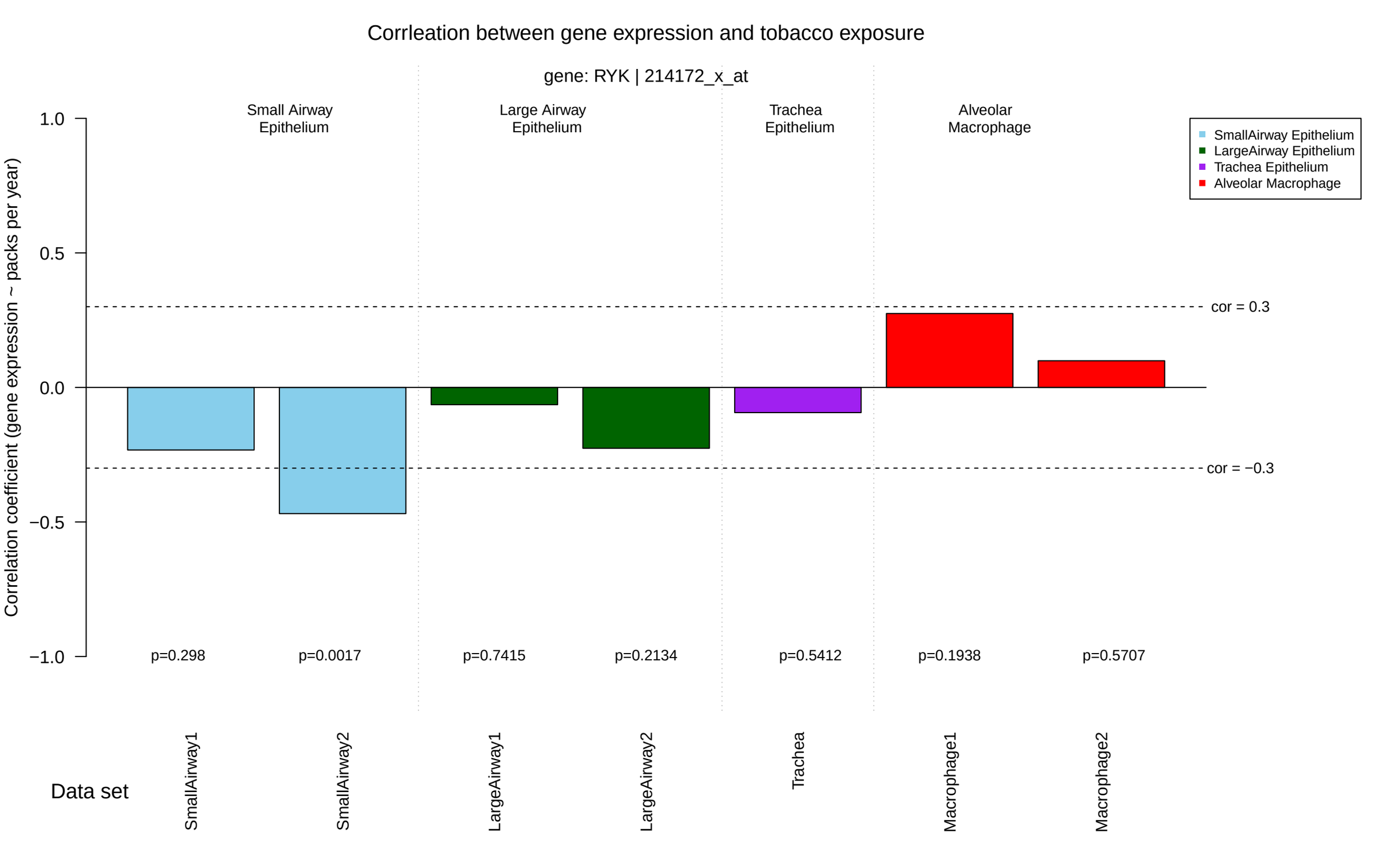

set: LargeAirway1 (p=0.051)
RYK | 214172_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.062)
RYK | 214172_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3815)
RYK | 214172_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0109)
RYK | 214172_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0699)
RYK | 214172_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RYK | 214172_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.298
p=0.0017
p=0.7415
p=0.2134
p=0.5412
p=0.1938
p=0.5707
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2